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Palaeomagnetism of volcanic rocks of the Kodiak Islands indicates northward latitudinal displacement

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In the past decade it has become widely accepted that southern Alaska is composed of allochthonous terranes[1–4], but the times of amalgamation and their accretion to North America remain controversial. Palaeomagnetists and geologists generally agree that the Talkeetna superterrane, consisting of the Wrangellia and Peninsular terranes (Fig. 1), was far to the south of its present position with respect to North America in Triassic and Jurassic time[1–4]. This consensus is lost when it comes to the question of the relative position of these terranes in the Tertiary. Previously, limited palaeomagnetic evidence suggested that the Peninsular terrane may have lain substantially south of its present position[4], but the timing of deformational and thermal events and the apparent absence of a Tertiary suture zone argue for a Cretaceous docking of Wrangellia and the Peninsular terrane[5,6]. Here we present new palaeomagnetic results demonstrating that the Kodiak Islands were indeed at a more southerly latitude 62 Myr ago, around 43° N, and discuss the implications of this conclusion.

We sampled the volcanic rocks of the Palaeocene Ghost Rocks Formation exposed along the southeastern margin of the Kodiak Islands (Fig. 1). This turbidite sequence consists of a landward belt with intercalated flows of andesite pillows and a seaward belt including basalt flows. The sedimentary rocks are commonly isoclinally folded and locally grade into zones of tectonic disruption. The volcanic rocks are generally much less deformed and typically form coherent folds. The Ghost Rocks Formation is part of a late Cretaceous to Palaeogene accretionary sequence. The formation is apparently thrust to the north-west against a late Cretaceous trench turbidite sequence[7] and underthrust from the south-east by an Eocene to Oligocene submarine fan complex[8]. Metamorphic prehnite, pumpellyite and laumontite are locally developed[9]. The youngest foraminifera (Palaeocene) and 62 Myr radiometric ages of cross-cutting plutons suggest an early Palaeocene depositional age for the Ghost Rocks Formation.

We drilled and oriented cores of Ghost Rocks volcanics from two regions on Kodiak Island—Kiliuda Bay and Alitak Bay—which are separated by ~80 km along strike. In all, an average of seven cores were collected from each of 29 flows in the two regions. Flow attitudes were almost always obtained from interbedded sediments, and only occasionally from pillow morphology. The Kiliuda Bay sites included locations in both the andesite belt and the basalt belt, whereas the Alitak Bay sites included flows from the andesite belt only.

Stepwise thermal demagnetization proved effective in cleaning the samples from almost all of the flows. Characteristic components of natural remanent magnetization were isolated at temperatures of 300 °C and higher that trended towards the origin on vector diagrams. Considerable improvement in clustering of directions resulted for most flows: the average k increased from 15 to 40 during thermal cleaning. Only 2 of the 29 sites failed to clean up, and one of these was a highly deformed, badly fractured tuff.

Both polarities were found in the flows collected at Kiliuda Bay. Application of the F-test shows that normal and reverse mean directions are not significantly different from antiparallel. These directions also pass the fold test: when correction is made for the tectonic dip of the flows, k increases from 4 in geographical coordinates to 29 in stratigraphical coordinates (Table 1). At Alitak Bay only reversely magnetized flows are present, but these also pass the fold test (Table 1). Thus there is strong evidence that the characteristic magnetization of the rocks pre-dates folding in both areas and probably preserves the original field direction. Even after tectonic correction, however, the mean directions for the two regions are very different (Table 1). The mean declinations differ by 97°, suggesting the regions may have been rotated with respect to each other about a vertical axis. The inclinations also differ from each other by a much smaller, but still statistically significant amount. A likely explanation for this is that the Alitak Bay group of flows spans insufficient time to average out secular variation, although other explanations involving systematic differences in initial dips or in ages of flows between the two regions cannot be categorically ruled out.

The most important palaeomagnetic parameter for regional-scale tectonic interpretation is the mean inclination of the Palaeocene geomagnetic field for the Kodiak Islands. It is reasonable to assume that some sort of average of the data sets for the two regions would give a better estimate of the true inclination than either one of them alone. There are several techniques that have been developed for averaging inclinations only, when declination data are unavailable or should be excluded because of scatter (refs 10, 11, and A. Cox and R. Gordon, in preparation). A straightforward and simple way to do it for this case, however, is to rotate the sets of directions for both regions so that their mean declinations coincide (Fig. 2). Note that the resulting distribution of directions shown in Fig. 2 does not appear particularly non-fisherian (certainly it is not bimodal), which supports the idea that the discrepancy in inclinations between the two regions may be due to incomplete averaging of secular variation. Analysis of the combined

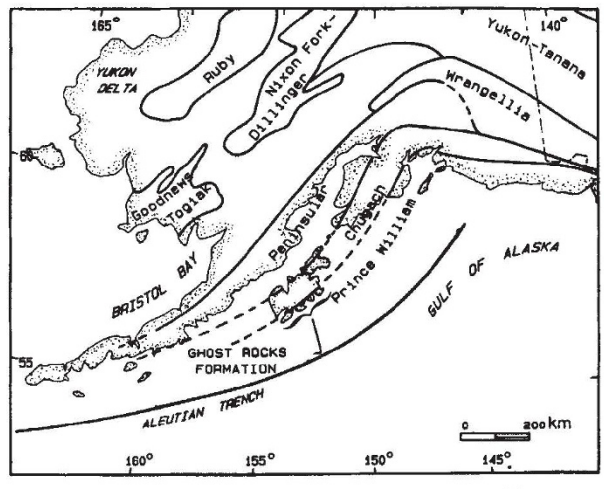


Fig. 1 Generalized terrane map of southern Alaska[14] showing location of the Ghost Rocks Formation.

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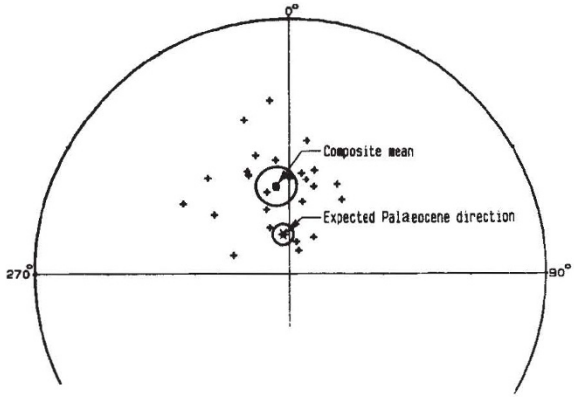


Fig. 2 Site mean directions (reversed inverted to normal): the mean declination for each region (Kiliuda Bay, Alitak Bay) has been rotated (31° clockwise, and 65° anticlockwide respectively) to coincide with the expected Palaeocene declination. Mean of composite data set (■) and expected Palaeocene direction (★) are shown with associated 95% confidence limits.

data set by Fisher statistics[12] gives a mean inclination of 61.4° and 95% confidence limits of ±6.5°. The same mean inclination, within a few tenths of a degree, is obtained by any of the more sophisticated inclination or colatitude statistics referred to above.

As shown in Fig. 2, the mean inclination of the Ghost Rocks volcanics is significantly shallower than that expected for Kodiak if it had been in its present position with respect to North America in Palaeocene time. The corresponding palaeolatitude is 43 ± 9° N, which is significantly less than the expected value of 66 ± 6° N derived from Palaeocene palaeomagnetic data for cratonic North America[13]. Thus the latitudinal displacement of Kodiak with respect to North America that is indicated is 23 ± 11° northward (Fig. 3); any longitudinal displacement is, of course, indeterminate.

The lithology, structural style and geological setting of the Ghost Rocks Formation suggest that it is a Palaeocene subduction complex. On the tectonostratigraphic map of Jones *et al.*[14] (Fig. 1) it is classified as part of the Prince William terrane. However, it is tied to the adjacent Chugach terrane (Fig. 1) by plutons of identical age (60–62 Myr) and composition. The only published palaeomagnetic data from the Chugach terrane are from two Eocene layered-gabbro plutons and a Cretaceous sheeted dyke and pillow basalt complex studied by Grommé and Hillhouse[15]. Although no fold test was described, their results from both sites suggest the Chugach terrane was about 25° south of its present position with respect to North America at these times, in excellent agreement with our results from the Prince William terrane.

The next terrane inboard is the Peninsular terrane (Fig. 1). It is simplest to suppose that the Peninsular terrane formed the backstop against which the late Cretaceous–early Tertiary accretionary complex of the Chugach–Prince William terranes formed, although the geological evidence for this is equivocal.

Table 1 Palaeomagnetic results for Ghost Rocks volcanics

	Dec	Inc	A95	*k*	*N*
Kiliuda Bay					
Geographical	345.5	61.9	25.9	4.1	11*
Stratigraphical	320.1	52.7	8.5	29.5	11*
Alitak Bay					
Geographical	168.2	36.0	12.6	10.9	14†
Stratigraphical	56.7	68.3	8.6	22.6	14†
Ghost Rocks					
Mean	351.4	61.4	6.5	21.3	25

* Two sites eliminated because of unstable magnetization.
† Two superposed flows with identical NMR eliminated because directions were grossly discordant (>2 s.d. from mean).

Certainly very similar accretionary processes are occurring there today. The relatively few Tertiary palaeomagnetic data that have been published for the Peninsular terrane are consistent with this idea. In a study of 74 cores of lower Tertiary sediments from four closely spaced localities near Pavlov Bay, Stone and Packer[4] found shallow inclinations that imply low palaeolatitudes comparable to those we have found for the Ghost Rocks Formation.

Much further inboard, just north of the Denali fault, one finally encounters concordant palaeomagnetic directions. In a study of the Teklanika volcanics, Hillhouse and Grommé[16] have found steep inclinations consistent with the expected palaeocene field. Our own work on Ordovician limestones and a few early Tertiary lava flows in the Nixon Fork terrane 80 km to the west is consistent with their finding[17]. Thus, according to the palaeomagnetic evidence, considerable Tertiary convergence must have been taken up somewhere south of the Denali fault.

Figure 3 shows the band of possible Palaeocene positions for the southern terranes compatible with the palaeomagnetic results for the Ghost Rocks Formation. We have chosen our study to position the southern terranes in the early Tertiary because it is the only one on igneous rocks of well determined age and palaeohorizontal control that passes fold and reversal tests. The locations of the Pacific, Farallon and North American plate boundaries are essentially those of Engebretson (in preparation), who assumes fixed hotspots. The position of the Kula-Farallon ridge is less well constrained because the relevant magnetic anomalies have been subducted.

We favour a Palaeocene position of the southern terranes near to the Kula–Farallon ridge as drawn in Fig. 3 because other evidence suggests that the Ghost Rocks volcanics were the product of a ridge–trench encounter. These lavas are much too close to the subduction zone to be normal arc volcanism[18,19], and their composition, as well as that of the correlative pluton, suggests that they all have a mid-ocean ridge basalt parent[20,21]. Positioning the southern terranes north of the Kula-Farallon ridge allows them to be pushed northward by the movement of the Kula and Pacific plates, rather than being diverted to

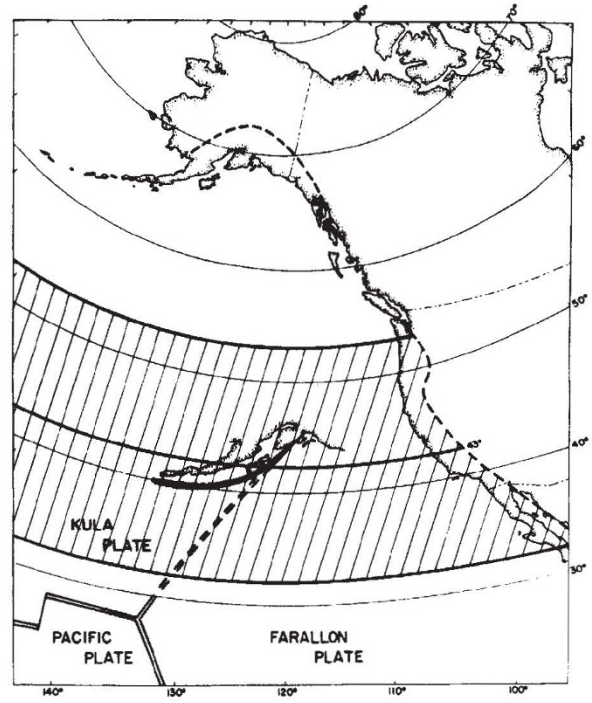


Fig. 3 Palaeocene reconstructiion showing palaeomagnetically inferred latitude for Ghost Rocks Formation, Kodiak, Alaska. Stippled palaeolatitudinal band is the uncertainty with respect to North America at the 95% confidence level. Heavy dashed lines suggest the Palaeocene North American plate margin.

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the east by the motion of the Farallon plate. The mystery of the missing Tertiary suture zone in central Alaska required by this reconstruction, however, remains an enigma. These ideas and the data on which they are based will be presented in full detail elsewhere.

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$^{40}Ar/^{39}Ar$ dating of pyrite

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The precise determination of the age of sulphide ore deposits is one of the most difficult problems in geochronology. This is largely because the very process responsible for concentrating certain metals to an economically significant level also tends to exclude radioactive isotopes from the deposits. It is customary to try to determine the age of mineralization by analysing silicate material presumed to be cogenetic[1]. Because such materials are not always available, and because it is not always clear that the silicates and ore minerals are in fact contemporaneous, we examined the sulphide minerals themselves to see if they contained sufficient potassium and argon for $^{40}Ar/^{39}Ar$ age determination. Our initial results indicate that this is the case for pyrite from the Geco ore body in northwestern Ontario, Canada.

The direct dating of sulphides with the Rb–Sr technique was attempted by Reesman and Hurley[2]. The results showed promise, but the approach was never pursued. More recently, Luck and Allègre[3] have shown that some deposits may be datable with the Re–Os method. Most promising is the pioneering work on the Rb–Sr dating of fluid inclusions in quartz from a tungsten deposit by Shepherd and Darbyshire[4].

Six pyrite samples have been analysed from the Geco mine, a stratabound and stratiform volcanogenic massive Cu–Zn sulphide deposit[5]. Geco has long been of interest to students of lead isotopes in connection with calculations of the age of the Earth and the chemical evolution of the mantle and crust[6]. The ore body has experienced the Kenoran orogeny[5] and therefore we had *a priori* information on its age. U–Pb and Rb–Sr dating in the area had shown[7] that the last severe regional metamorphism occurred 2,600–2,700 Myr ago. Furthermore, we could add a much more specific age constraint to the ore body when, during mineral separation, we discovered minute amounts of the easily datable biotite in intimate association with sphalerite crystals. A few flecks of these micas were therefore hand-picked for age determination along with the six pyrite samples.

All samples were irradiated in the McMaster University reactor along with the Zartman hornblende standard[8]. Step-heating runs were performed on two biotite samples whilst the pyrites, because of their extremely low potassium concentrations (5–35 p.p.m.), were fused in single step analyses. The mass spectrometric isotope analysis of the evolved argon was carried out with an MS10 machine. The results are shown in Figs 1 and 2 and Tables 1 and 2. The biotite age spectra in Fig. 1 show excellent plateaus and there is evidence of only minute argon loss in the low temperature fractions. The plateaus are identical within experimental error and indicate that the cooling age of the ore body, after the Kenoran orogeny, is 2,610 ± 20 Myr. The age spectra indicate that the deposit has undergone negligible heating since then.

The pyrite data are shown in Fig. 2 where $^{40}Ar/^{36}Ar$ is plotted against $^{39}Ar/^{36}Ar$. Such a graphical analysis must, of course, be used as there is no knowing *a priori* what the $^{40}Ar/^{36}Ar$ composition was in the pyrites at the time they cooled below their argon retention temperatures. The data points in Fig. 2 are clearly strongly linearly correlated, but the scatter about a straight line exceeds that due solely to expected experimental error. The scatter cannot be accounted for by system blank

Table 1 Analytical data for Geco biotites (step-heating)

T (°C)	Atmospheric (%)	$^{40}Ar^*$ ($\times10^{-5}$ cm^3 STP g^{-1})	$f(39)$	Age (Myr)
Sample GB250†				
810	1.2	6.21	3.7	2,530 ± 31
910	0.2	42.80	24.3	2,589 ± 3
1,100	0.2	55.51	30.8	2,620 ± 3
1,600	0.2	73.54	41.3	2,604 ± 2
Total	0.2	178.06	Integrated age	2,604 ± 5
Standard $(40/39)_s = 43.57 \pm 0.19$				
Mass = 0.0091g				
Sample GB1050				
550	52.8	0.08	0.2	1,137 ± 142
650	24.8	0.17	0.5	802 ± 104
700	5.9	0.88	0.5	2.426 ± 41
780	2.3	2.78	1.4	2,614 ± 8
820	0.4	21.94	11.5	2,600 ± 5
950	0.1	73.51	37.4	2,637 ± 3
1,170	0.1	89.78	45.8	2,633 ± 4
1,600	2.0	5.21	2.7	2,630 ± 11
Total	0.3	194.35	Integrated age	2,623 ± 6
Standard $(40/39)_s = 45.22 \pm 0.32$				
Mass = 0.0197g				

Decay constants from ref. 11 were used. Errors are $\pm1\sigma$.
† Only negligible volumes of $^{40}Ar^*$ were released from GB250 below 800 °C.

Table 2 Analytical data for Geco pyrites

Sample	Mass (g)	K (p.p.m.)	$^{40}Ar^*$ ($\times10^{-7}$ cm^3 STP g^{-1})	$^{40}Ar/^{36}Ar$	$^{39}Ar/^{36}Ar$
OGS-10	0.2478	5.6	1.97	393.3 ± 4.5	0.34 ± 0.02
OGS-14	0.2613	8.8	2.56	444.4 ± 2.8	0.54 ± 0.01
OGS-131	0.8568	34.4	7.34	1,219.0 ± 9.0	5.13 ± 0.04
OGS-132	0.8484	16.8	3.72	840.9 ± 15.5	2.88 ± 0.02
OGS-133	0.8933	4.8	2.76	631.1 ± 7.6	0.68 ± 0.01
OGS-134	0.9348	11.9	3.06	521.4 ± 6.3	0.56 ± 0.01

Errors are $\pm1\sigma$. $^{40}Ar/^{36}Ar$ and $^{39}Ar/^{36}Ar$ values have been corrected for a system blank of 1×10^{-8} cm^3 STP. The system blank has atmospheric argon composition.

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